OPTIMIZATION – MODEL ANSWERS

1. a) The stationary points of the function f are computed by solving the equations

$$0 = \nabla f = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2(x_1-2)(2(x_1-2)^2+x_2^2) \\ 2x_1^2x_2 - 8x_1x_2 + 10x_2 + 2 \end{bmatrix}.$$

As a result, the point $x_\star = (2,-1)$ is the unique stationary point. [2 marks]

Typical mistakes include the incorrect solution of the equations defining the stationary points or the identification of more than one solution. Note that imaginary solutions should be discarded.

b) The Hessian matrix of the function f is

$$\nabla^2 f(x) = \begin{bmatrix} 12(x_1-2)^2+2x_2^2 & 4(x_1-2)x_2 \\ 4(x_1-2)x_2 & 2(x_1-2)^2+2 \end{bmatrix},$$

hence

$$\nabla^2 f(x_\star) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

Since $\nabla^2 f(x_\star) > 0$, $x_\star$ is a minimizer of f. Note now that

$$0 \leq (x_1-2)^4 + (x_2+1)^2 \leq f,$$

and the function $(x_1-2)^4+(x_2+1)^2$ takes non-negative values and it is radially unbounded (it is the sum of two squares, one involving x_1 and one involving x_2). Hence, f is radially unbounded, and since $f(x_\star) = 0$, $x_\star$ is the global minimizer of f. [4 marks]

Typical mistakes include writing a non-symmetric Hessian matrix, checking positivity of the Hessian matrix by checking only its determinant, inferring that the function is convex using the Hessian at the stationary point only, and not providing a convincing argument that the function is radially unbouded.

c) The modified Newton's iteration is given by

$$x_{k+1} = x_k - \frac{1}{2}\nabla f(x_k) = \begin{bmatrix} x_{1,k} - (x_{1,k}-2)(2(x_{1,k}-2)^2+x_{2,k}^2) \\ -4x_{2,k} - x_{2,k}x_{1,k}^2 + 4x_{2,k}x_{1,k} - 1 \end{bmatrix}.$$

[4 marks]

The most common mistake involves the use of the standard Newton's iteration, instead of the modified one, as given in the text of the exam paper. This generates very complex formulas, which have been incorrectly derived.

d) The points generated by the modified Newton's iteration from the starting point $x_0 = (3/2, 0)$ are

$$x_1 = (1.75, -1),\ x_2 = (2.03125, -0.9375),\ x_3 = (2.003723145, -.9990844731),$$

Answers

$x_4 = (2.000006711, -.9999861507), \qquad x_5 = (2.000000000, -1.000000000).$

[4 marks]

Most mistakes are numerical mistakes in the computation of the sequence.

e) The points generated by the modified Newton's iteration from the starting point $x_0 = (1,0)$ are

$$x_1 = (3,-1), \qquad x_2 = (0,0), \qquad x_3 = (16,-1), \qquad x_4 = (-5486,195).$$

[2 marks]

Most mistakes are numerical mistakes in the computation of the sequence.

f) The research direction used in the modified Newton's iteration is $-1/2\nabla f(x_k)$, which is nothing else than the direction of the anti-gradient, hence it is a descent direction satisfying the condition of angle. The reason why the method is not globally convergent is that the line search parameter is *fixed* to $\alpha = 1$, and this may not yield a descent algorithm at each step. [4 marks]

Most of the students failed to recognize that the modified Newton method is nothing else than the gradient method with line search parameter fixed to $1/2$. In addition, most students have been unable to provide a reasonable argument for the local convergence and have simply stated that this is expected since Newton method is locally convergent.

Answers

2. a) Note that

$$\nabla f = \begin{bmatrix} x_1 \\ m x_2 \end{bmatrix},$$

hence the gradient algorithm is described by the iteration

$$x_{1,k+1} = x_{1,k} - \alpha x_{1,k}, \qquad x_{2,k+1} = x_{2,k} - \alpha m\, x_{2,k}.$$

Replacing x_{k+1} in f yields

$$f(x_{k+1}) = \frac{1}{2}\left(x_{1,k}^2 + m\, x_{2,k}^2\right) - \alpha\left(x_{1,k}^2 + m^2 x_{2,k}^2\right) + \frac{1}{2}\left(x_{1,k}^2 + m^3\, x_{2,k}^2\right)\alpha^2.$$

To obtain the exact linear search parameter one has to compute the stationary point of $f(x_{k+1})$ as a function of α (since $f(x_{k+1})$ is convex in α), that is

$$\alpha_\star = \frac{x_{1,k}^2 + m^2 x_{2,k}^2}{x_{1,k}^2 + m^3\, x_{2,k}^2}.$$

As a result, the gradient algorithm with exact line search is given by

$$x_{k+1} = x_k - \alpha_\star \nabla f(x_k),$$

as given in the exam question. [6 marks]

The typical mistakes include the inability to write the gradient iteration for the function as a function of the line search parameter and then overlooking that this is a function of one parameter, α, with respect to which the function has to be minimized. The solution of this optimization problem provides the required iteration.

b) As indicated in the exam question, for the considered initial condition and value of m the value of $\alpha_\star$ is constant, namely

$$\alpha_\star = \frac{x_{1,0}^2 + m^2 x_{2,0}^2}{x_{1,0}^2 + m^3\, x_{2,0}^2} = 1/5.$$

As a result, the gradient iteration is given by

$$x_{1,k+1} = \frac{4}{5} x_{1,k}, \qquad x_{2,k+1} = -\frac{4}{5} x_{2,k}.$$

This yields

$$x_{1,k} = x_{1,0}\left(\frac{4}{5}\right)^k = 9\left(\frac{4}{5}\right)^k, \qquad x_{2,k} = x_{2,0}\left(-\frac{4}{5}\right)^k = (-1)^k\left(\frac{4}{5}\right)^k,$$

as indicated in the exam paper. [8 marks]

The students have failed to exploit the fact that for the given initial conditions the optimal line search parameter is constant for all iterations. Then they have been unable to see that the iteration yields nothing else than a discrete-time linear systems for which solutions are directly determined.

c) Note that $x_\star = 0$, hence

$$\|x_{k+1}\|^2 = \left(9\left(\frac{4}{5}\right)^{k+1}\right)^2 + \left((-1)^{k+1}\left(\frac{4}{5}\right)^{k+1}\right)^2 = 82\left(\frac{4}{5}\right)^{2(k+1)},$$

Answers

$$\|x_k\|^2 = 82\left(\frac{4}{5}\right)^{2k},$$

thus

$$\frac{\|x_{k+1} - x_\star\|}{\|x_k - x_\star\|} = \frac{4}{5}.$$

The sequence thus converges with linear speed of convergence. [6 marks]

Mistakes include the incorrect computation of the error and of the relative error. In addition, the students have failed to identify the correct speed of convergence (that is, linear) and have argued (occasionally in strong terms) for incorrect answers.

Answers

3. a) To begin with, rewrite the inequality constraint as $4 - x_1 - x_2 \leq 0$. The Lagrangian of the problem is

$$L(x_1, x_2, \rho) = 2x_1^2 + 9x_2 + \rho(4 - x_1 - x_2).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x_1} = 4x_1 - \rho, \qquad 0 = \frac{\partial L}{\partial x_2} = 9 - \rho,$$

$$\rho \geq 0, \qquad (4 - x_1 - x_2) \leq 0, \qquad \rho(4 - x_1 - x_2) = 0.$$

[4 marks]

Typical mistakes include writing only some of the necessary conditions or not transforming the problem to a standard formulation. This had then an impact on the direction of some inequalities.

b) Note that $\rho = 9$, hence the constrain has to be satisfied with the equality sign, thus yielding the only candidate solution $x_{1,\star} = 9/4$, $x_{2,\star} = 7/4$, $\rho = 9$.

[4 marks]

Multiple, incorrect, candidate solutions may have been computed or the complementarity condition may have been used incorrectly.

c) The stationary points of B_r are obtained solving the equations

$$0 = \nabla B_r = \begin{bmatrix} 4x_1 - \dfrac{r}{(4 - x_1 - x_2)^2} \\ 9 - \dfrac{r}{(4 - x_1 - x_2)^2} \end{bmatrix}.$$

The second equation yields $\dfrac{r}{(4 - x_1 - x_2)^2} = 9$, which substituted in the first equation yields $x_1 = 9/4$. Replacing $x_1 = 9/4$ in

$$(4 - x_1 - x_2)^2 = \frac{r}{9}$$

and solving for x_2 yields two stationary points

$$P_1 = \left(\frac{9}{4}, \frac{7}{4} + \frac{1}{3}\sqrt{r}\right), \qquad P_2 = \left(\frac{9}{4}, \frac{7}{4} - \frac{1}{3}\sqrt{r}\right).$$

The point P_1 is admissible for all $r > 0$, whereas P_2 is outside the admissible set. Note also that

$$\lim_{r \to 0} P_1 = \left(\frac{9}{4}, \frac{7}{4}\right) = P_{1,\star},$$

which coincides with the candidate optimal solution determined in part b).

[6 marks]

Mistakes include the incorrect differentiation of the barrier function, the incorrect solution of the necessary conditions of optimality, and the computation of only one stationary point (instead of two). In addition, students have failed to check admissibility of the approximate solutions or to verify the "limit" condition.

Answers

d) The necessary conditions of optimality for the constrained problem are

$$0 = \frac{\partial L}{\partial x_1}, \qquad 0 = \frac{\partial L}{\partial x_2},$$

and those for B_r are

$$0 = \frac{\partial B_r}{\partial x_1}, \qquad 0 = \frac{\partial B_r}{\partial x_2}.$$

Comparing the conditions yields

$$\frac{\partial L}{\partial x_1} = \frac{\partial B_r}{\partial x_1}, \qquad \frac{\partial L}{\partial x_2} = \frac{\partial B_r}{\partial x_2}.$$

As a result

$$\rho = \frac{r}{(4 - x_1 - x_2)^2},$$

which evaluated at P_1 yields $\rho = 9$, consistently with part b). [6 marks]

This is a sort of "conceptual" question: the students had to see two different objects and identify similarities. The computations were also non-trivial because of the need to compute a limit and some students have overlooked this.

Answers

4. a) The Lagrangian of the problem is

$$L(x_1, x_2, \lambda) = 2(x_1^2 + x_2^2 - 1) - x_1 + \lambda(x_1^2 + x_2^2 - 1).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x_1} = 4x_1 - 1 + 2\lambda x_1, \qquad 0 = \frac{\partial L}{\partial x_2} = 4x_2 + 2\lambda x_2, \qquad 0 = x_1^2 + x_2^2 - 1.$$

[2 marks]

Mistakes include the incorrect derivation of the necessary conditions of optimality or the incorrect definition of the Lagrangian.

b) The condition $0 = \frac{\partial L}{\partial x_2}$ yields either $x_2 = 0$ or $\lambda = -2$. The latter, replaced in $0 = \frac{\partial L}{\partial x_1}$ yields $0 = -1$, hence no candidate solution. The former yields two candidate solutions

$$S_1 = (x_1, x_2, \lambda) = \left(1, 0, -\frac{3}{2}\right), \qquad S_2 = (x_1, x_2, \lambda) = \left(-1, 0, -\frac{5}{2}\right).$$

[4 marks]

The student may have computed only one candidate optimal solution, or more than two solutions, or may have associated the wrong multiplier to a specific solution.

c) Note that

$$\nabla^2 L(S_1) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \nabla^2 L(S_2) = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

hence S_1 yields a local minimizer, that is the solution of the problem. Note that S_2 yields a local maximizer. [4 marks]

Mistakes include the incorrect computation of the Hessian (for example a non-symmetric Hessian) and the incorrect use of the complete sufficient conditions, which are not needed since the Hessian is a multiple of the identity matrix. In addition, the use of incorrect multipliers may have caused an incorrect classification of the stationary points.

d) The point $x_k = (\cos\theta_k, \sin\theta_k)$ is admissible.

i) The linearization of the constraint is given by the (linear) constraint

$$2(\cos\theta_k x_1 + \sin\theta_k x_2 - 1) = 0.$$

[2 marks]

Mistakes include the incorrect linearization of the constrain and overlooking that the considered parameterization implies admissibility of the point x_k.

ii) The Lagrangian of the problem is

$$L_l(x_1, x_2, \lambda) = 2(x_1^2 + x_2^2 - 1) - x_1 + 2\lambda(\cos\theta_k x_1 + \sin\theta_k x_2 - 1).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L_l}{\partial x_1} = 4x_1 - 1 + 2\lambda\cos\theta_k, \qquad 0 = \frac{\partial L_l}{\partial x_2} = 4x_2 + 2\lambda\sin\theta_k,$$

Answers

$$0 = \cos\theta_k x_1 + \sin\theta_k x_2 - 1.$$

These are linear equations in x_1, x_2 and λ with the unique solution

$$x_{1,\star} = \cos\theta_k + \frac{1}{4}\sin^2\theta_k, \qquad x_{2,\star} = \sin\theta_k - \frac{1}{4}\sin\theta_k\ \cos\theta_k,$$

with multiplier $\lambda = -2 + 1/2\cos\theta_k$. [4 marks]

This is a computationally intensive question in which the students have to manipulate "big objects'. Most mistakes are related to an incorrect definition of the problem or of the Lagrangian. The necessary conditions may have also been incorrectly solved. Note that the unique solution should have been parameterized in θ_k and not all students have seen this.

iii) The update law is given by

$$\theta_{k+1} = \frac{\sin\theta_k(4-\cos\theta_k)}{4\cos\theta_k + \sin^2\theta_k}.$$

The sequence generated from the initial value $\theta_0 = 0.1$ is

$$\theta_1 = 0.07518803294, \quad \theta_2 = 0.05647085060, \quad \theta_3 = 0.04238693407,$$

$$\theta_4 = 0.03180448680, \qquad \theta_5 = 0.02385939890.$$

From the above one could conclude that the sequence converges to $\theta = 0$, however it takes more than five iterations to increase by one the number of correct digits. This slow convergence is typical of algorithms with linear speed of convergence. [4 marks]

Some students have failed to note that the iteration is in terms of θ (not x_1 or x_2). There were also several numerical mistakes in the computation of the sequence θ_i and some students have failed to recognize that this was a sequence converging with linear speed.

Answers